Difference of Means

October 5, 2022

As you come in...

Load in the ANES data we used last class (if this code doesn't work, replace the arrow with an equals sign):

```
anes< -read.csv("https://raw.githubusercontent.com/
Brandon-Bolte/PLSC309_F22/main/ANES.csv")
```

The media has become polarized, but NPR is still widely accepted as a neutral news source. I want to know if watching NPR (variable is "npr") is related to whether people voted for Trump ("trump2016") and Clinton ("clinton2016").

1. Following the steps we went through last class, formulate a hypothesis test to answer this question. You can choose to examine Trump or Clinton voters. Write down (or type) each step (but don't worry about the critical value)
2. Hint: Chi square test command is chisq.test()

Next Week...

Bivariate Hypothesis Testing

	Independent Variable		
Dependent Variable		*Categorical*	*Continuous*
	Categorical	Crosstab/Chi-square	Logit/Probit
	Continuous	Difference of Means/T-test; Bivariate Regression	Correlation Coefficient; Bivariate Regression

Formulating a Hypothesis Test

1. State question
2. State the null hypothesis
3. State the alternative hypothesis
4. State the desired confidence level
5. Find critical value
6. Calculate test statistic
7. Compare test statistic to critical value and state conclusion (can we reject the null?)

Bivariate t-test: Difference of Means

- One categorical variable and one continuous variable
- Is the mean of the continuous variable in one category the same as the mean in the other category?
  - In other words, are they plausibly drawn from the same underlying distribution in the population?
  - In other other words, is the difference between the means 0?

Bivariate t-test: Difference of Means

- One categorical variable and one continuous variable
- Is the mean of the continuous variable in one category the same as the mean in the other category?
  - In other words, are they plausibly drawn from the same underlying distribution in the population?
  - In other other words, is the difference between the means 0?

Null hypothesis: the sample means are drawn from the same underlying distribution (e.g., difference in means is 0... means are basically the same)

Alternative hypothesis: the sample means are not drawn from the same underlying distribution (e.g., the means are very different from each other, very much not 0)

T-test Statistic

$$t = \frac{\bar{Y}_1 - \bar{Y}_2}{se(\bar{Y}_1 - \bar{Y}_2)} \sim t_{n_1+n_2-2} \tag{1}$$

- $\bar{Y}_i$ = the mean of each group i
- n_i = size of each group i
- $se(\bar{Y}_1 - \bar{Y}_2)$ = standard error of difference of means

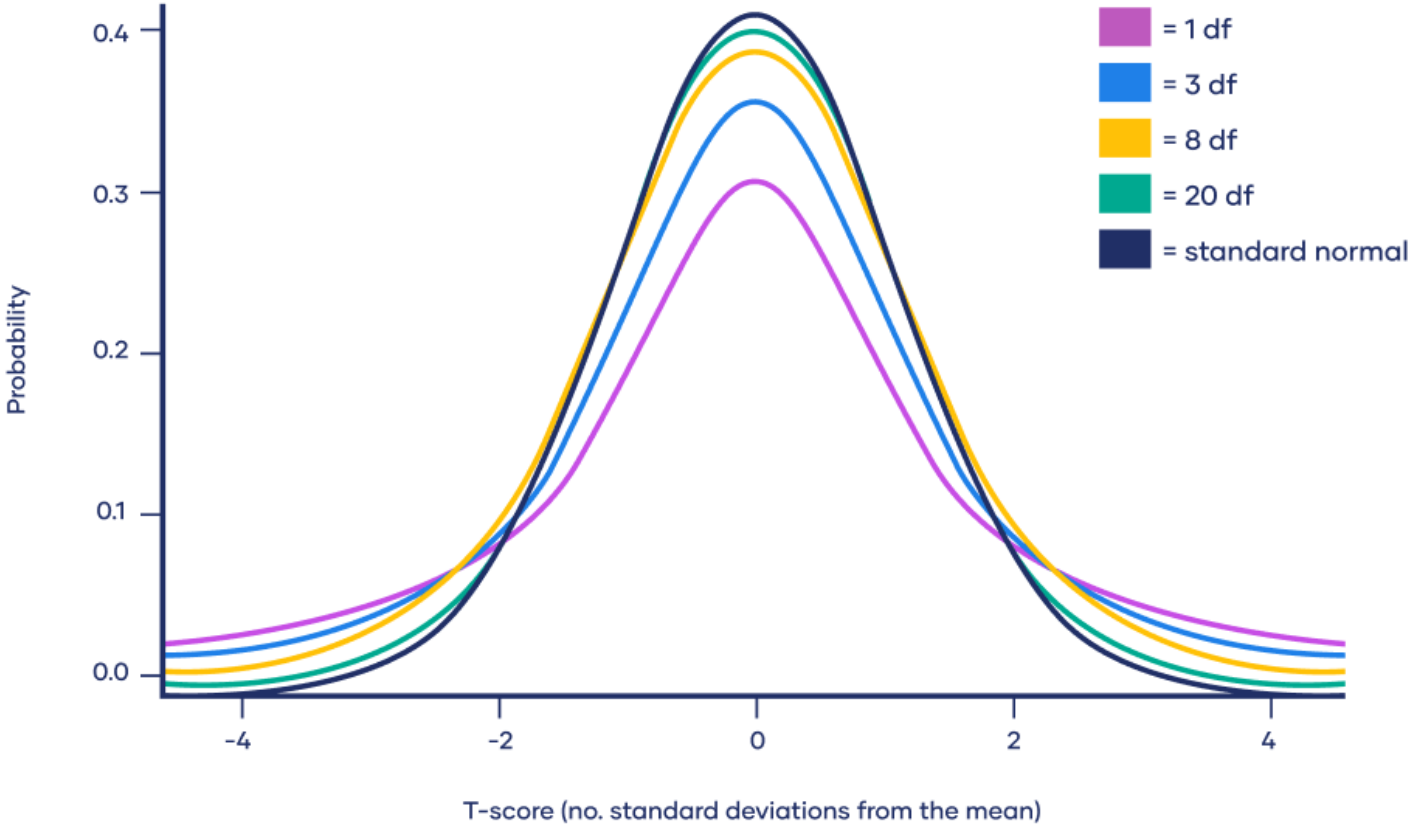
Standard Error of Difference of Means

NOT the difference of the two standard errors. Much scarier to calculate:

$$se(\bar{Y}_1 - \bar{Y}_2) = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n + 2 - 2}} \times \sqrt{(\frac{1}{n_1} + \frac{1}{n_2})} \quad (2)$$

- n_i is the sample size in group i
- s_i^2 is the variance of group i

The t-distribution



One-sample t-test: $df = n - 1$

Two-sample t-test: $df = n_1 + n_2 - 2$

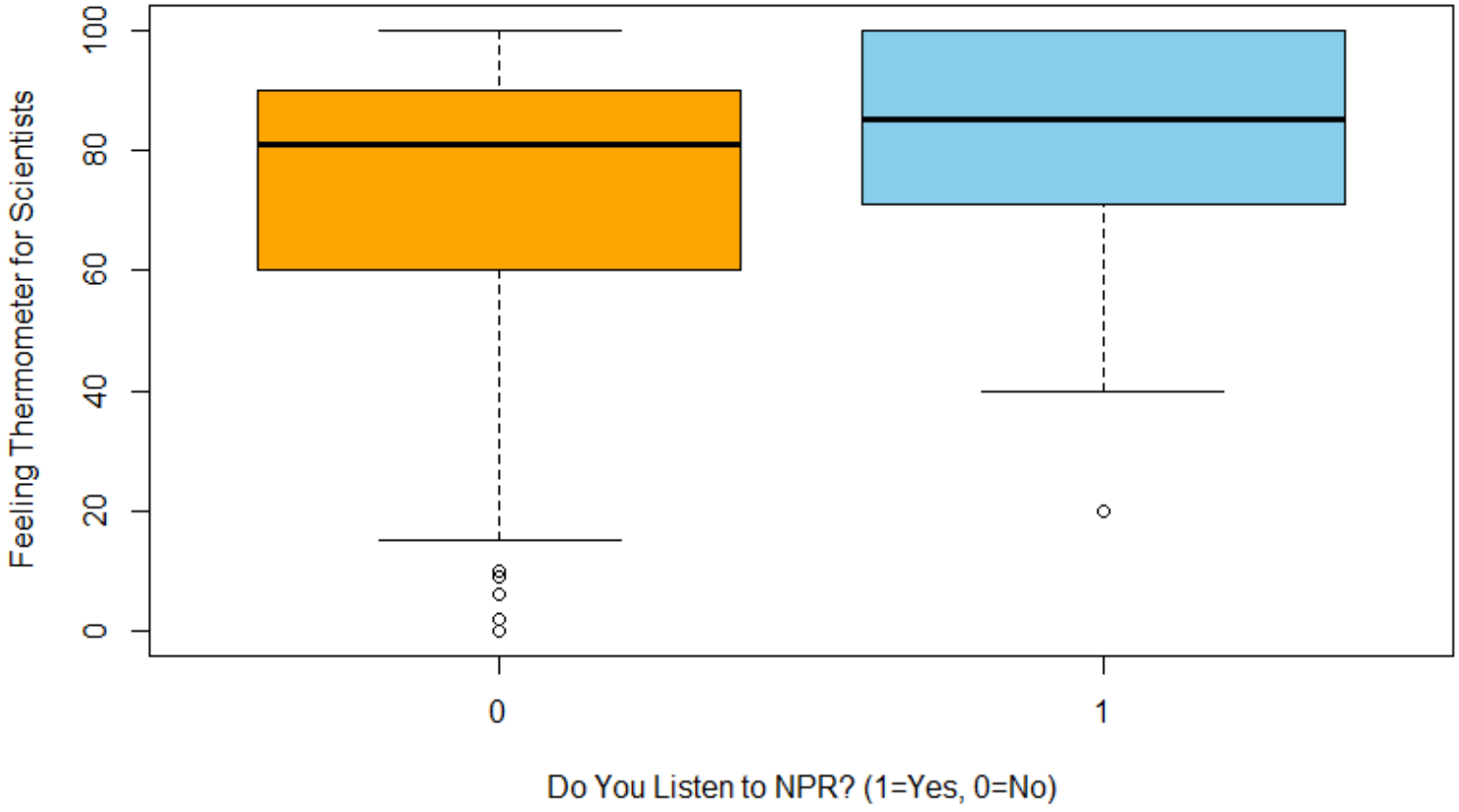
Example: NPR Viewership and Scientist Favorability

Research Question: Is there a difference in favorability for scientists among NPR listeners compared to non-NPR listeners?

- Independent Variable: Do you listen to NPR? (Yes/No)
- Dependent Variable: Feeling Thermometer for scientists (0-100)

Logic of the test: Is the mean feeling thermometer score for scientists the same for NPR listeners and non-listeners? In other words, is the difference between these means basically zero?

Feeling Thermometer for Scientists by NPR Listeners



Mean FT for NPR Listeners: 85.62

Mean FT for non-NPR Listeners: 75.79

Formulating a Hypothesis Test

1. State question
2. State the null hypothesis
3. State the alternative hypothesis
4. State the desired confidence level
5. Find critical value
6. Calculate test statistic
7. Compare test statistic to critical value and state conclusion (can we reject the null?)

Formulating a Hypothesis Test

1. **Is there a difference in favorability for scientists among NPR listeners compared to non-NPR listeners?**
2. State the null hypothesis
3. State the alternative hypothesis
4. State the desired confidence level
5. Find critical value
6. Calculate test statistic
7. Compare test statistic to critical value and state conclusion (can we reject the null?)

Formulating a Hypothesis Test

1. **Is there a difference in favorability for scientists among NPR listeners compared to non-NPR listeners?**
2. **Null: There is no difference between the means**
3. State the alternative hypothesis
4. State the desired confidence level
5. Find critical value
6. Calculate test statistic
7. Compare test statistic to critical value and state conclusion (can we reject the null?)

Formulating a Hypothesis Test

1. **Is there a difference in favorability for scientists among NPR listeners compared to non-NPR listeners?**
2. **Null: There is no difference between the means**
3. **Alternative: There is a difference between the means**
4. State the desired confidence level
5. Find critical value
6. Calculate test statistic
7. Compare test statistic to critical value and state conclusion (can we reject the null?)

Formulating a Hypothesis Test

1. **Is there a difference in favorability for scientists among NPR listeners compared to non-NPR listeners?**
2. **Null: There is no difference between the means**
3. **Alternative: There is a difference between the means**
4. **I want to test my hypothesis at the 95% confidence level**
5. Find critical value
6. Calculate test statistic
7. Compare test statistic to critical value and state conclusion (can we reject the null?)

Find Critical Value Step 1: Degrees of Freedom

DF for T-distribution:

$$n_1 + n_2 - 2$$

In our data:

- $n_1 = 298$
- $n_2 = 1{,}674$
- So, df = 298 + 1,674 - 2 = 1,970

	Level of significance					
df	0.10	0.05	0.025	0.01	0.005	0.001
1	3.078	6.314	12.706	31.821	63.657	318.313
2	1.886	2.920	4.303	6.965	9.925	22.327
3	1.638	2.353	3.182	4.541	5.841	10.215
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.893
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.782
8	1.397	1.860	2.306	2.896	3.355	4.499
9	1.383	1.833	2.262	2.821	3.250	4.296
10	1.372	1.812	2.228	2.764	3.169	4.143
11	1.363	1.796	2.201	2.718	3.106	4.024
12	1.356	1.782	2.179	2.681	3.055	3.929
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
20	1.325	1.725	2.086	2.528	2.845	3.552
25	1.316	1.708	2.060	2.485	2.787	3.450
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
50	1.299	1.676	2.009	2.403	2.678	3.261
60	1.296	1.671	2.000	2.390	2.660	3.232
70	1.294	1.667	1.994	2.381	2.648	3.211
75	1.293	1.665	1.992	2.377	2.643	3.202
80	1.292	1.664	1.990	2.374	2.639	3.195
90	1.291	1.662	1.987	2.368	2.632	3.183
100	1.290	1.660	1.984	2.364	2.626	3.174
∞	1.282	1.645	1.960	2.326	2.576	3.090

Formulating a Hypothesis Test

1. **Is there a difference in favorability for scientists among NPR listeners compared to non-NPR listeners?**
2. **Null: There is no difference between the means**
3. **Alternative: There is a difference between the means**
4. **I want to test my hypothesis at the 95% confidence level**
5. **My degrees of freedom are more than 100, so we estimate that our critical value is 1.645**
6. Calculate test statistic
7. Compare test statistic to critical value and state conclusion (can we reject the null?)

T-test Statistic

$$t = \frac{\bar{Y}_1 - \bar{Y}_2}{se(\bar{Y}_1 - \bar{Y}_2)} \sim t_{n_1+n_2-2}$$

$$t = \frac{85.62 - 75.79}{se(\bar{Y}_1 - \bar{Y}_2)} = \text{- } \mathbf{10.048}$$

Note: I won't ask you to calculate the denominator...

Formulating a Hypothesis Test

1. **Is there a difference in favorability for scientists among NPR listeners compared to non-NPR listeners?**
2. **Null: There is no difference between the means**
3. **Alternative: There is a difference between the means**
4. **I want to test my hypothesis at the 95% confidence level**
5. **My degrees of freedom are more than 100, so we estimate that our critical value is 1.645**
6. **My test statistic is 10.048**
7. Compare test statistic to critical value and state conclusion (can we reject the null?)

Formulating a Hypothesis Test

1. **Is there a difference in favorability for scientists among NPR listeners compared to non-NPR listeners?**
2. **Null: There is no difference between the means**
3. **Alternative: There is a difference between the means**
4. **I want to test my hypothesis at the 95% confidence level**
5. **My degrees of freedom are more than 100, so we estimate that our critical value is 1.645**
6. **My test statistic is 10.048**
7. **Since 10.048 is greater than 1.645, I can reject the null hypothesis that the means are the same at the 95% confidence level**

In R...

t.test(**Continuous Variable** ~Grouping Variable)

```
> t.test(anes$scientistsft~anes$npr)

        Welch Two Sample t-test

data:  anes$scientistsft by anes$npr
t = -10.048, df = 500.37, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -11.748291  -7.905506
sample estimates:
mean in group 0 mean in group 1
       75.79391        85.62081
```

Final Notes about T-tests

- Remember, we use the t-distribution for one-sample tests and bivariate tests between a nominal and continuous variable

Final Notes about T-tests

- Remember, we use the t-distribution for one-sample tests and bivariate tests between a nominal and continuous variable
- Unlike Chi-square, technically T-distribution can be positive or negative

Final Notes about T-tests

- Remember, we use the t-distribution for one-sample tests and bivariate tests between a nominal and continuous variable
- Unlike Chi-square, technically T-distribution can be positive or negative
- Technically there are two types of T-test (Student's T and Welch's T)
  - Book teaches Student's T, R defaults to Welch's T
  - For our purposes, don't worry about the difference, but might see slight differences if you do by hand vs in R